Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)

1 Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)

Study focus: Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 5 on energy storage devices and we'll discuss about fuel cells.

Key details

- In today's class, we are going to learn about hydrogen oxygen alkaline fuel cell.
- The principle, construction and working, the advantages and the disadvantages.
- Alkaline fuel cells as the name suggests, the electrolyte which we are going to use is going to be an alkali.
- If you remember when we talked about fuel cells in general, we use a fuel at the anode and an oxidant at the cathode, and there is an electrolyte.
- These are low temperature fuel cells and they operate at about 100 degrees centigrade.
- Most common oxid which is used at the cathode is oxygen.
- Oxygen reduction is a very slow process.
- But if we use alkaline electrolytes,.
- As compared to acid electrolytes, it is found that the oxygen reduction is much more rapid in alkaline electrolytes.
- And because of that, we need not use noble metal catalysts.
- Oxygen reduction or hydrogen oxidation.
- Hydrogen oxidation which happens at the anode or oxygen reduction which happens at the cathode being slow processes we need to use catalysts.
- But because it is faster, the oxygen reduction is faster in the alkaline electrolyte as compared to acid electrolyte, we can use non noble metal electrocatalysts.

- One problem with alkaline fuel cells is that carbon containing fuels cannot be used as when they undergo oxidation, carbon dioxide is formed as a product.
- This K2CO3 is going to block the electrodes, the electrode pathways, and therefore this reduces the efficiency of the cell.
- The reaction would be carbon dioxide, which is formed by the due to the oxidation of carbon containing fuels with KOH is going to give you K2 CO3 plus H two O.
- used and has many applications and the electrolyte obviously is going to be an alkali.
- At the anode it will be hydrogen which is going to be used.
- At the cathode we will use oxygen and the electrolyte is going to be an alkali.
- The oxidant which we use is oxygen.
- and the electrolyte is KOH that is 30 to 45% of KOH is used.
- But one thing we need to note here that the KOH used has to be warm.
- We'll discuss the reason in the upcoming slides.
- These are the electroactive materials.
- Hydrogen and oxygen are the electroactive materials, but we also need electrodes.

Example or application from the lecture

- Let us discuss the construction of such a fuel cell.
- Now we will take the example of hydrogen oxygen fuel cell, which is the most common type of alkaline fuel cell.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let us look at the diagram.

Key details

- Hydrogen because hydrogen has to undergo oxidation and at the anode the process is always oxidation.
- Carbon electrode is impregnated with silver.
- One compartment is the anodic compartment where we let in the fuel.
- And we also have a exit for the excess fuel and on the cathode side that is the other compartment where we have the cathode here is the entry for air or oxygen in this case and then there is also an exit point for the excess air or oxygen.
- and in the center we have the electrolyte that is the KOH which is being used.
- And apart from that we have the electrodes which are shown here that is the anode and this side is the cathode.
- When we say working, we are basically going to understand the reactions which are taking place at the anode and cathode and the overall reaction.
- And then at the cathode, these electrons are getting absorbed.
- Overall reaction when you add up these two reactions the half cell reactions they are known as the anode and the cathode, we find that it is hydrogen plus half O2 is giving you water.

- But what we have done is we have separated the oxidation and the reduction reaction into separate compartments.
- These electrons move through the external wire and they are taken up at the cathode because of the reduction of oxygen happening here.
- So this way by the separating the reactions into anode and cathode just like we do for any galvanic cell, we have been able to have the stream of electrons flowing and therefore we can generate power from this cell.
- Hence, it is very important that the anode has to be porous in nature.

Example or application from the lecture

- Let us understand the working of such a cell.
- OH minus and it is going to form water and two electrons are given out here because it's a oxidation reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Expedite the reaction.

Key details

- In some cells we find that there is a vent provided over here and from here the water escapes.
- Whenever we make any fuel cells it is very essential that the electrodes have to be porous in nature because they have to absorb the fuel the oxidant and also the catalyst is present on the electrode.
- So therefore the hydrogen oxygen is getting ads at the cathode surface and it gets reduced to OH minus ions.
- Therefore, what we do is we operate the cell at 100 degrees so that the water which is formed from KOH can be it is going to escape as steam.
- Because this water which is getting formed is obviously going to dilute the KH and we have used a particular concentration of KH.
- And if the KVH gets very diluted, then the efficiency of the reaction is going to fall down.
- Therefore, we make sure that this excess water.
- Which is getting formed continuously is removed from the system.
- Hence, it is very important that water is allowed to go out of the system, and that is why we operate the cell at about 100 degrees centigrade, or in other words,.
- Why we use warm KH so that water escapes as steam and we maintain the concentration of KVH which is required for that for the reaction.
- And one more interesting fact which you must be familiar with is that during the Apollos phase program the water which was coming out of this fuel cell was collected and the astronauts were using this water for drinking.
- Now the EMF which is generated because we are using this fuel cell basically to generate power.
- Coming to advantages of this fuel cell, it operates at a low temperature.

- So because of that, corrosion of equipment which is being used in the fuel cell happens to a very small extent as compared to those fuel cells which are operate at a very high temperature.
- And alkali which is used as the electrolyte is allowing us to use a non-noble metal catalyst and because of that what happens is the cell is now less expensive.
- So we can afford to use non-mobile metal catalysts here and therefore the it will work out to be much less expensive as compared to other fuel cells.
- And there are certain disadvantages that the reactants must be free from carbon.
- We also saw this point that if we have carbon containing fuels, then they will undergo oxidation to form carbon dioxide, and this reacts with alkali.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

And it forms carbonates, which is going to reduce the efficiency of the cell.

Key details

- which have which are used has to be free of carbon compounds.
- And other problem is that since we are using electrolyte which is a liquid, then that could pose some handling problems.
- And depending on the application where we are going to use this fuel cell, we will decide which type of fuel cell is suitable or whether this fuel cell is suitable for that particular application.
- So in the next class we are going to talk about methanol oxygen fuel cell where we are going to use a polymer membrane as the electrolyte.

Example or application from the lecture

- So it also means that the fuel and the oxidant, suppose we are talking of hydrogen and oxygen, then fuel and oxidant.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9972; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.

4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

C Revision and retrieval practice